Digital Attendance Monitoring System: A Console-Based Python Application for Student Attendance Recording and Management

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Introduction To Computing Project
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PRESENTED BY

CABADING, JUSTIN JAY S.
CLEOPAS, ADONIS L. JR.
MANTICA, ARNZT FINLEY A.
PABATAO, FLIANE DENVER S.
REYES, FEBE AMOR L,
URO, REINEL JOHN C.

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1.1 Introduction

Attendance is one of the first tasks handled in many classes, yet small mistakes during this routine can create extra work later. A teacher may call names while students answer, write marks on a class list, or check a sign-in sheet before the lesson continues. When the class is large or time is limited, a name can be skipped, a status can be marked in the wrong row, or the same student can appear more than once. A student who notices a mistake may need to ask the teacher to check the record, while the teacher must look back at the list and confirm what was entered. These situations are simple, everyday classroom concerns, but they show why attendance information should be arranged in a way that is easy to review and correct.

Attendance records help instructors keep track of class participation and review patterns of absence. In a study of more than 500 undergraduates, Martínez-Serna et al. (2024) found that higher attendance was associated with a greater likelihood of taking and passing the exam and with higher final grades, though the authors note their results show correlation rather than causation. Attendance alone does not determine a student's academic performance, but such findings help explain why maintaining clear attendance records is an important part of classroom management.

Classroom attendance is commonly recorded through teacher-led roll call or student sign-in, and automated alternatives have also been developed. Ishaq and Bibi (2023) point out that manual methods consume class time, allow proxy signing, and risk lost attendance sheets. Ali et al. (2022) reviewed 90 studies on attendance management systems, covering their advantages, methods, and obstacles, while Nguyen et al. (2022) described systems built on RFID, fingerprint, QR code, and facial recognition. These studies show that digital tools may help organize the process, but the approach still needs to match the users' needs and the resources available.

With these considerations, the proponents propose the Digital Attendance Monitoring System, a menu-driven Python program for basic attendance recording and checking in two class sections, COC-FC2-BSIT1-01 AND COC-FC2-BSIT1-02. When the program starts, users can choose the Student or Teacher option or exit the program. Students choose their section, enter their student ID and name, and the program records the current date and marks the student Present. Teachers choose a section, set the total number of students in that section, view the entries for a chosen date, check the total, Present, and Absent counts, add a missing record, correct a selected record, or delete an incorrect or duplicate entry. The program aims to organize these basic attendance tasks while demonstrating programming concepts that the proponents can realistically develop and explain.

1.2 Statement of the Problem

General Problem

The general problem addressed by this project is how to make routine attendance recording and checking more organized through a simple Python program. In a manual process, a teacher may need to record each response, recount the marks, and look back through the class list when a student questions an entry. When details are incomplete, incorrect, or duplicated, correcting the record can take additional time. The proposed system will provide a menu for student entry and teacher review, with basic options to add, correct, and delete records during the current program session for two class sections.

Specific Problem

The project focuses on three practical problems that can occur during regular classroom attendance:

- Overcrowding and Disorganized Flow

During roll call or sign-in, an instructor may need to record each student's response while keeping the class moving. Afterward, the teacher may still need to organize the entries or count the number of students present and absent. A simple menu-based program can keep the submitted entries together during the current session and display them for checking.

- Incomplete, incorrect or duplicate entries

Attendance information can be entered with a missing student ID, misspelled name, wrong date, or incorrect status. A student's details may also be entered more than once. If the record is not checked, the teacher may have to spend time finding the mistake and deciding which entry to keep. The proposed program will check that the student ID follows the required number format, that the name is not empty, that the date follows the YYYY-MM-DD format, and that the status is Present or Absent. It will also refuse a second entry for the same student ID on the same date, in either section, while giving the teacher an option to update an entry or delete an incorrect or duplicate record. These checks can help organize the list, but they cannot prove that the information entered is true.

- Difficulty in reviewing records and attendance total

When a teacher needs to confirm a student's status or count the students marked Present and Absent, searching a handwritten list and recounting the marks can take extra effort. The program will display numbered attendance entries for a chosen section and date and basic totals (the total number of students, the number Present, and the number Absent)

1.3 General Objectives

The main objective of this project is to develop and test a simple, menu-driven Python attendance monitoring program that allows students to submit attendance details

and enables teachers to set the total of a chosen section and view, add, correct, and delete attendance entries for two class sections during the current program session.

1.4 Specific Objectives

Specifically, the proponents aim to:

- Create a role-selection menu that allows users to choose Student, Teacher, or Exit.
- Let users choose between sections COC-FC2-BSIT1-01 and COC-FC2-BSIT1-02, keeping the records and totals of each section separate.
- Allow students to choose their section and enter their student ID and name, with checks that the ID follows the required number format and that the name is not empty, while the program records the current date and marks the student Present.
- Allow teachers to choose a section, set its total number of students, view the attendance entries of that section for a chosen date, and review the total, Present, and Absent counts.
- Provide teacher options to add a missing record, correct a selected field in an attendance record or delete an incorrect or duplicate entry.
- Test the program using sample entries, including an invalid student ID, an empty name, a duplicate entry, a class that is already complete, the same ID in both sections, invalid status values, record updates, record deletion, and attendance totals.

Scope & Limitation

1.5 Scope

The Digital Attendance Monitoring System is a beginner-level, console-based Python program designed to organize basic student attendance recording and teacher review for two class sections, COC-FC2-BSIT1-01 and COC-FC2-BSIT1-02. When the program

opens, users choose whether to continue as a Student or Teacher, or to exit. After choosing Student, the user picks a section from a list. The student menu accepts a student ID and a name. The program checks that the ID uses numbers only in the format 02-2425-002996 and that the name is not empty. It then records the current date, marks the student Present in the chosen section, and adds the entry to a Python list. It does not add a second entry for the same ID on the same date, and it stops new Present entries once the number of Present students in that section equals the total set by the teacher. The teacher menu first asks the teacher to choose a section. The teacher can then set the total number of students of that section and display the numbered entries of that section for a chosen date (today by default). It shows the total students, the number Present, and the number Absent, which is computed as the total minus the number Present. On the same screen, the teacher can add a record marked Present or Absent, correct a selected field, or delete an incorrect or duplicate entry after confirming. The records and totals of the two sections are kept separate.

The program is intended to demonstrate introductory computing concepts, including variables, lists, input and output, conditional statements, loops, functions, input validation, and basic counting. The proponents will test these functions with sample data to check whether entries can be added, displayed, updated, deleted, and counted as intended. Students and teachers must use the same running program on the same computer to view the same in-session records. A Wi-Fi or local network connection may be available in the intended school environment, but network-based sharing between separate devices is not part of the current prototype.

1.6 Limitation

The program relies on information entered manually. It does not independently verify a student's identity or confirm whether the student is physically present in class.

The Student option is a role-selection menu, not a secure login; the program does not use passwords or authenticate the person choosing the Teacher option. For this reason, the teacher must review the entries and should not treat the program alone as proof of attendance. The program checks that the ID and the date follow the required number patterns, but it does not check that a date is a real calendar date or that an ID belongs to an enrolled student. Student entries use the date of the computer running the program, so that date must be correct. The Absent count also depends on the total number of students entered by the teacher for that section.

Attendance entries are stored in a Python list only while the program is running. They are lost when the program closes because the project does not use a database or permanent file storage. The teacher can add, correct, or delete records during the current session, but those changes are not saved for a later session. The total number of students set for each section is also lost when the program closes. The program does not include separate user accounts, secure access controls, QR scanning, automated notifications, permanent attendance history, or automatic tracking of tardiness and repeated absences. It does not connect multiple devices or synchronize records over a network.

The program is made for two class sections only, COC-FC2-BSIT1-01 and COC-FC2-BSIT1-02, which are written in the program. Other sections cannot be chosen, and adding one would require changing the section list in the program. The section is picked from a list, and the program does not check that the person really belongs to that section. A student ID can have only one entry per date across both sections. These limits keep the project within a realistic scope for an introductory Python course. It is intended to demonstrate basic attendance entry and record management, not to replace an official school attendance platform. If the system is later expected to let students submit attendance from their own devices while a teacher checks it on another device, additional network communication and a method of saving shared records would be needed. Those

functions should not be claimed as implemented unless they are developed and tested by the proponents.

1.7 System Requirements

Hardware Requirements

- Processor: Intel Core i3 or an equivalent processor.
- RAM: Minimum of 4 GB.
- Storage: At least 1 GB of available disk space for Python, Visual Studio Code, and project files.
- Monitor/Display: Resolution of 1024 × 768 or higher.
- Input Devices: Keyboard and mouse for entering information and operating the program.
- Network Connectivity: Wi-Fi or a local area network (LAN) connection is included for the intended school environment. The current prototype does not use the network to transfer or synchronize attendance records.
- Network Equipment: A Wi-Fi router or access point may be available in the school or laboratory environment; it is not used for record sharing by the current prototype.

Software Requirements

- Operating System: Windows 10 or Windows 11.
- Programming Language: Python 3.x.
- Development Environment: Visual Studio Code.

- Additional Libraries: No third-party libraries are required for the basic console functions described in this paper.

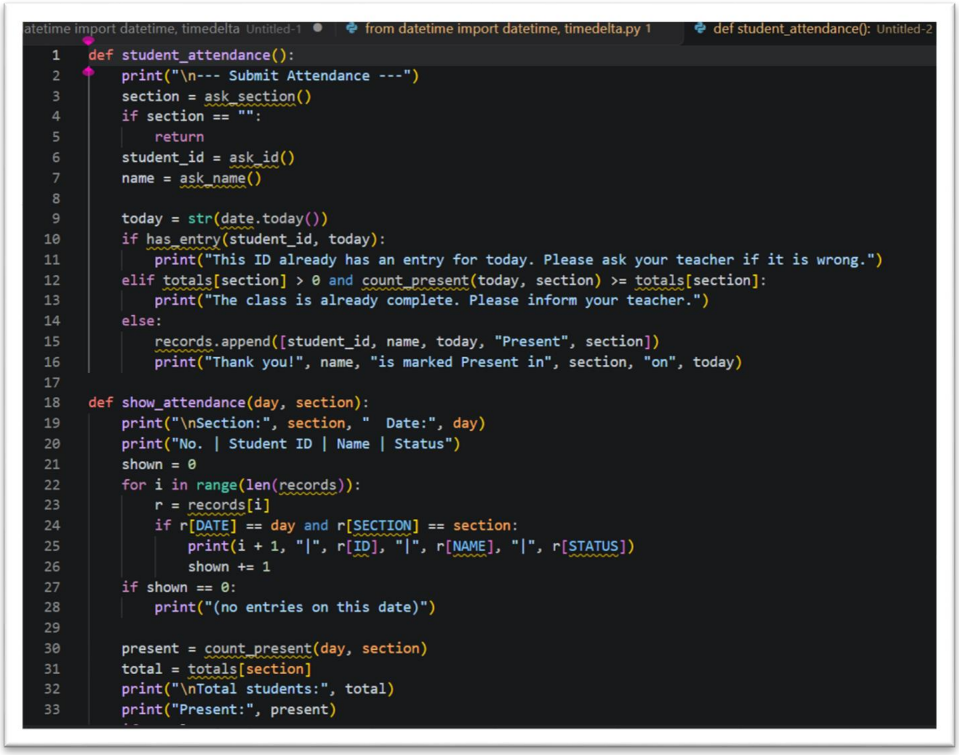
Student Functional Requirements (Primary User)

- Choose the Student option from the role-selection menu.
- Choose a section (COC-FC2-BSIT1-01 or COC-FC2-BSIT1-02), or press Enter to go back.
- Enter the required student ID (numbers only, such as 02-2425-002996) and name.
- Have the entry accepted and marked Present with the current date recorded automatically, only when the ID and name are valid, the ID has no entry yet for that date, and the chosen section is not already complete.
- Return to the role-selection menu after each submission or exit the program using the available menu options.

Teacher Functional Requirements (Secondary User)

- Choose the Teacher option from the role-selection menu.
- Choose a section after choosing the Teacher option.
- Set the total number of students of the chosen section (1 to 100).
- View all attendance entries of the chosen section for a chosen date (today by default), including entries marked Present and Absent.
- Review the displayed total students, Present, and Absent counts.
- Add a record for a student, with the same ID and name checks and a status of Present or Absent.
- Select an entry and correct its student ID, name, date, or attendance status, or delete an incorrect or duplicate entry after confirming.

1.8 Code Snippet of the Digital Attendance Monitoring System



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